

Sec 5.2 – The Definite Integral

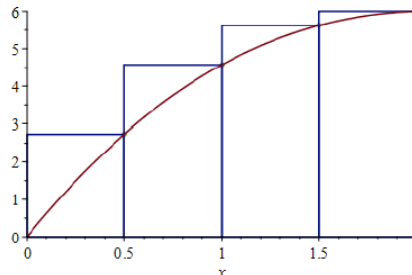
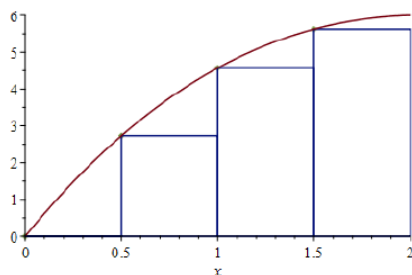
(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

In the previous section, we saw how a total value (in this case gallons of oil spilled) can be approximated by a sum of areas of rectangles. And we saw that this approximation improves as the width of the rectangles gets smaller.

In the rest of this chapter, we will consider more general functions, not just total gallons from gallons/hour.

Suppose $f(t)$ is a continuous function for $a \leq t \leq b$. We divide the interval from a to b into n equal subdivisions, and the width of each individual subdivisions is called Δt and calculated $\Delta t = \frac{b-a}{n}$.

$t_0, t_1, \dots, t_n$ are the endpoints of the subdivisions.



Left-hand sum:

$$f(t_0)\Delta t + f(t_1)\Delta t + f(t_2)\Delta t + \cdots + f(t_{n-1})\Delta t$$

or

$$\Delta t(f(t_0) + f(t_1) + f(t_2) + \cdots + f(t_{n-1}))$$

Right-hand sum

$$f(t_1)\Delta t + f(t_2)\Delta t + f(t_3)\Delta t \cdots + f(t_n)\Delta t.$$

We can factor out the Δt and the sum becomes: $\Delta t \cdot (f(t_0) + f(t_1) + \cdots + f(t_n))$.

A “nickname” for summations uses the symbol Σ , which is the Greek letter capital “S.” We sometimes refer to this as “Sigma” notation.

$$\text{Right-hand sum} = f(t_1)\Delta t + f(t_2)\Delta t + \cdots + f(t_n)\Delta t = \sum_{i=1}^n f(t_i)\Delta t.$$

The Σ tells us to add terms of the form $f(t_i)\Delta t$. The “ $i = 1$ ” at the base of the sigma sign tells us to start at $i = 1$, and the “ n ” at the top tells us to stop at $i = n$.

In the left-hand sum we start at $i = 0$ and stop at $i = n - 1$, so we write

$$\text{Left-hand sum} = f(t_0)\Delta t + f(t_1)\Delta t + \cdots + f(t_{n-1})\Delta t = \sum_{i=0}^{n-1} f(t_i)\Delta t.$$

Partner up

Let $f(t) = t^2$ and we want to estimate distance on the interval $[1, 9]$. Find Δt . Then write out the left and right sum for $n = 4$.

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With $n = 4$ intervals, $\Delta t = \frac{9-1}{4} = 2$.

Also, $t_0 = 1, t_1 = 3, t_2 = 5, t_3 = 7, t_4 = 9$. Then

Left-hand sum:

$$f(1) \cdot 2 + f(3) \cdot 2 + f(5) \cdot 2 + f(7) \cdot 2 = 1^2 \cdot 2 + 3^2 \cdot 2 + 5^2 \cdot 2 + 7^2 \cdot 2 = 168$$

or you can use the shortcut to compute:

$$2(f(1) + f(3) + f(5) + f(7)) = 2(1^2 + 3^2 + 5^2 + 7^2) = 168$$

Right-hand sum:

$$f(3) \cdot 2 + f(5) \cdot 2 + f(7) \cdot 2 + f(9) \cdot 2 = 3^2 \cdot 2 + 5^2 \cdot 2 + 7^2 \cdot 2 + 9^2 \cdot 2 = 328$$

or you can use the shortcut to compute:

$$2(f(3) + f(5) + f(7) + f(9)) = 2(3^2 + 5^2 + 7^2 + 9^2) = 328$$

Suppose we still let $f(t) = t^2$ and we want to estimate distance on the interval $[1, 9]$. How can we get a better approximation that we did on the last slide, where $n = 4$?

MORE INTERVALS! So write out the left sum for $n = 100$...

Note that using Sigma is much more compact as n gets larger:

Right sum:

$$f(t_1)\Delta t + f(t_2)\Delta t + f(t_3)\Delta t \cdots + f(t_n)\Delta t = \sum_{i=1}^n f(t_i)\Delta t$$

Left sum:

$$f(t_0)\Delta t + f(t_1)\Delta t + f(t_2)\Delta t + \cdots + f(t_{n-1})\Delta t = \sum_{i=0}^{n-1} f(t_i)\Delta t$$

And if we let n continue to grow, we will get a better approximation, and then as $n \rightarrow \infty$ we get the **exact** answer:

Suppose f is continuous for $a \leq t \leq b$. The **definite integral** of f from a to b , written

$$\int_a^b f(t) dt,$$

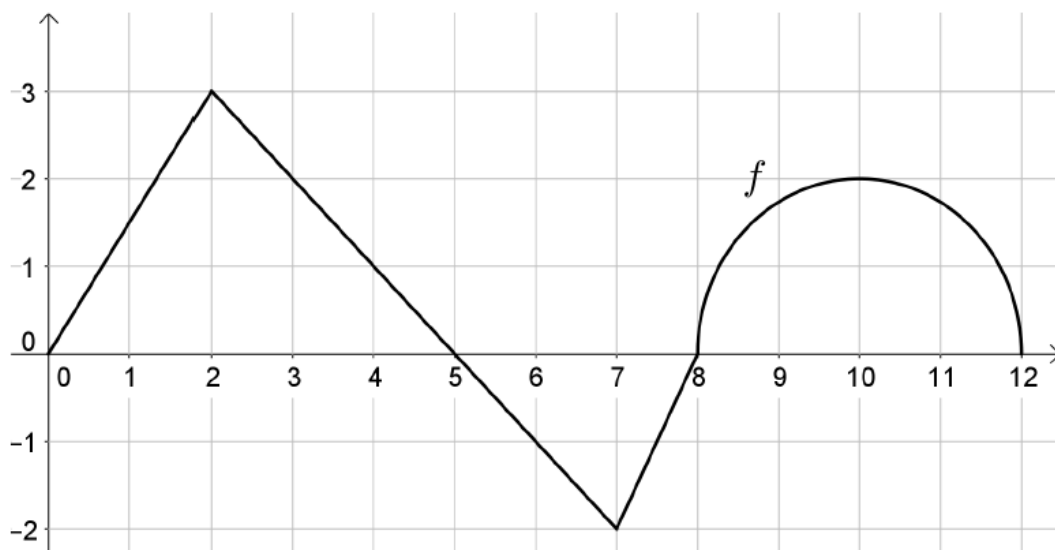
is the limit of the left-hand or right-hand sums with n subdivisions of $a \leq t \leq b$ as n gets arbitrarily large. In other words,

$$\int_a^b f(t) dt = \lim_{n \rightarrow \infty} (\text{Left-hand sum}) = \lim_{n \rightarrow \infty} \left(\sum_{i=0}^{n-1} f(t_i) \Delta t \right)$$

and

$$\int_a^b f(t) dt = \lim_{n \rightarrow \infty} (\text{Right-hand sum}) = \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n f(t_i) \Delta t \right)$$

Each of these sums is called a Riemann sum, f is called the integrand, and a and b are called the limits of integration.



Using the graph above, evaluate the definite integrals:

- $\int_0^2 f(x) dx$
- $\int_0^5 f(x) dx$
- $\int_5^8 f(x) dx$
- $\int_8^{12} f(x) dx$
- $\int_0^{12} f(x) dx$

- $\int_0^2 f(x) dx$

Solution: This is the area under f on the interval $[0, 2]$, which is a triangle with base 2 and height 3. So $\int_0^2 f(x) dx = \frac{1}{2}(2)(3) = 3$.

- $\int_0^5 f(x) dx$

Solution: This area is a triangle with base 5 and height 3, so $\int_0^5 f(x) dx = \frac{1}{2}(5)(3) = 7.5$.

- $\int_5^8 f(x) dx$

Solution: This area is the triangle under the x-axis and therefore the definite integral is negative. The triangle has base 3 and height 2, so

$$\int_5^8 f(x) dx = -\frac{1}{2}(3)(2) = -3.$$

- $\int_8^{12} f(x) dx$

Solution: This area is a semicircle, with radius 2. Therefore,

$$\int_8^{12} f(x) dx = \frac{1}{2}\pi(2)^2 = 2\pi.$$

- $\int_0^{12} f(x) dx$

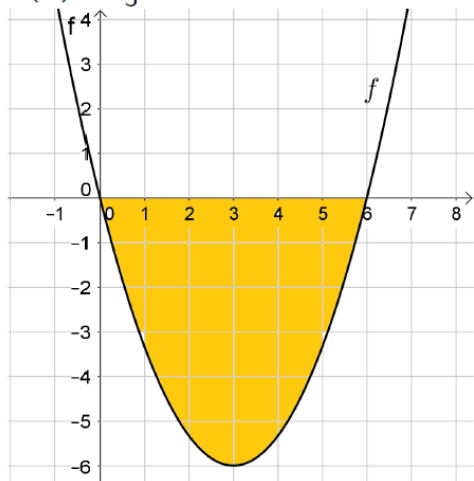
Solution: Here we put all the areas together:

$$\int_0^{12} f(x) dx = 7.5 - 3 + 2\pi = 4.5 + 2\pi.$$

When $f(x)$ is positive for some x values and negative for others, and $a < b$: $\int_a^b f(x) dx$ is the sum of areas above the x -axis, counted positively, and areas below the x -axis, counted negatively.

How can you find the area shaded below using a definite integral?

$$f(x) = \frac{2}{3}x^2 - 4x$$



$$\int_0^6 \left(\frac{2}{3}x^2 - 4x \right) dx$$

gives the *negative* area. CoCalc evaluates this signed area as -24 .

We can get the (unsigned) area by switching the sign: the area between the parabola and the x -axis is 24 square units.

Observations:

- If $f(t)$ is a *velocity* function, $\int_a^b f(t)dt$ will give us the *change in distance* or *displacement* from $t = a$ to $t = b$.
- We had noted that the change in distance was the *area* under the curve from $t = a$ to $t = b$.
- If we find the area exactly, we can find the value of the definite integral exactly. If we can't find the area exactly, we can approximate the area/distance in the same way we have already.
- Area above the x -axis counts positively toward the definite integral and area below the x -axis counts negatively toward the definite integral.

Time permitting:

Consider the function $f(x) = x^2 + 3$. We want to estimate the area under the function on $[2, 14]$ using a Riemann sum with 4 intervals.

- What is Δx ?
- Write out the terms $\sum_{i=1}^4 f(x_i)\Delta x$, where $x_0 = 2$.
- What integral is this estimating?

Solutions:

- $\Delta x = \frac{b-a}{n} = \frac{14-2}{4} = 3$
- Adding 3 each time, we get $x_0 = 2, x_1 = 5, x_2 = 8, x_3 = 11, x_4 = 14$.

$$\begin{aligned} \text{Then } \sum_{i=1}^4 f(x_i)\Delta x &= f(x_1)\Delta x + f(x_2)\Delta x + f(x_3)\Delta x + f(x_4)\Delta x \\ &= 3f(5) + 3f(8) + 3f(11) + 3f(14) \\ &= 3(28) + 3(67) + 3(124) + 3(199) = 1254 \end{aligned}$$

- This is a right-sum estimating $\int_2^{14} x^2 + 3 dx$